Olist customer retention: overview

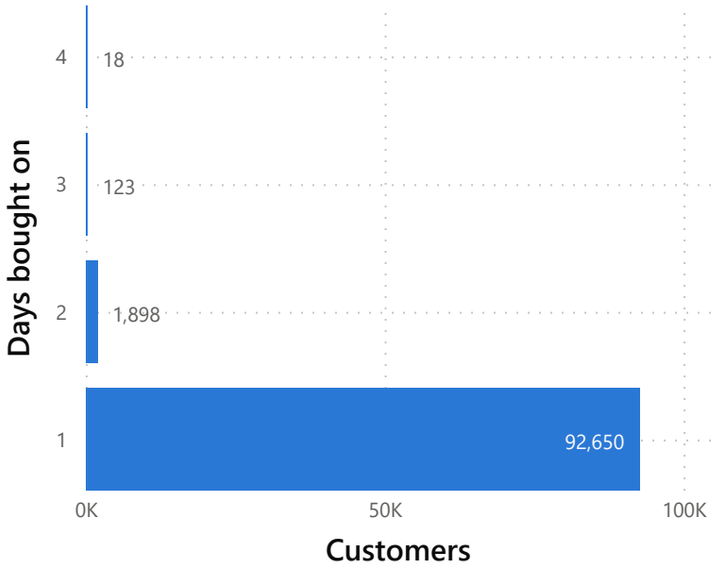
97.8% of customers buy once. Revenue rests on new customers.

94,703	R$15.68M	1.98%	4.21%
Customers	GMV	Repeat Rate 180d	Repeat Customer Revenue Share

Monthly GMV



Customers by number of days they bought on



State

All

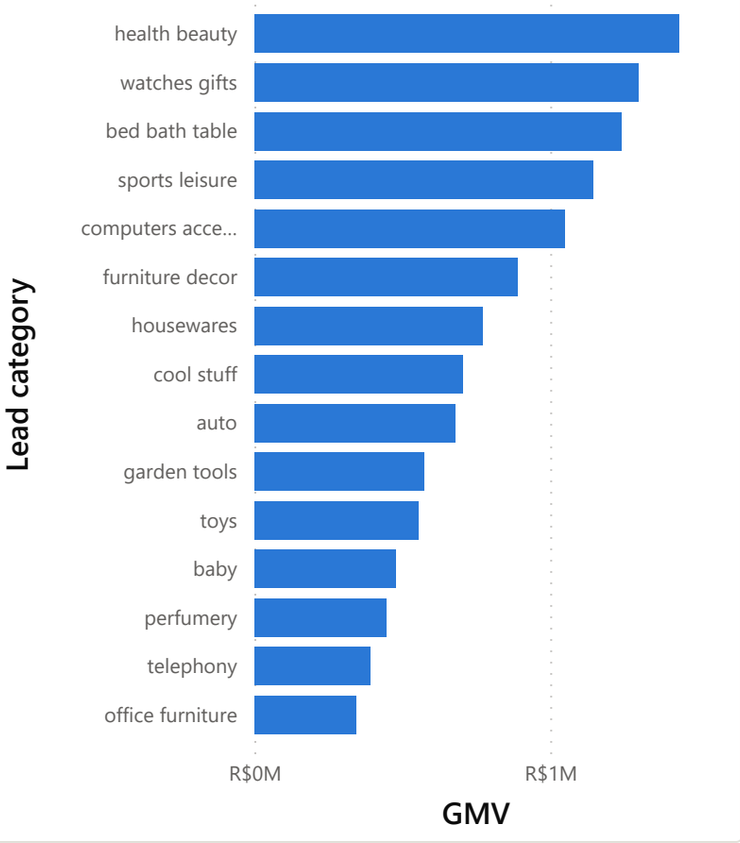
Source: Brazilian E-Commerce Public Dataset by Olist (Kaggle, CC BY-NC-SA 4.0).

Orders and geography

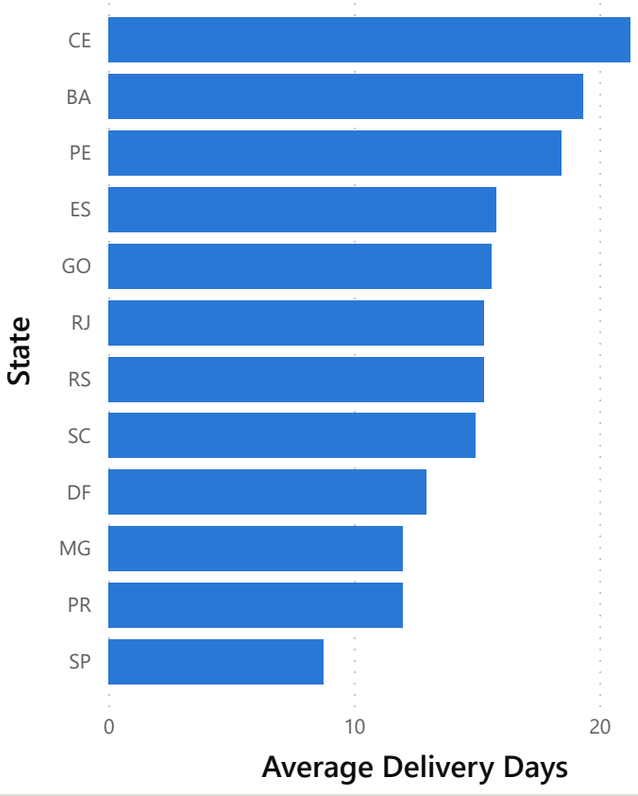
São Paulo state has 42% of customers and the fastest delivery.

97,905	R$160.19	12.5
Completed Orders	Average Order Value	Average Delivery Days

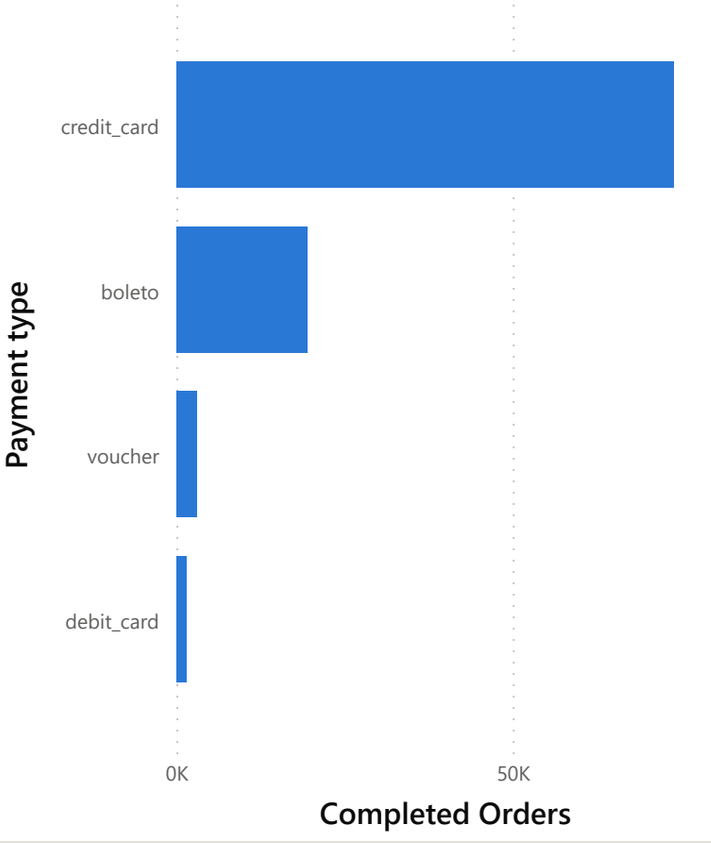
GMV by lead category (top 15)



Average delivery days by state (1,000+ orders)



Completed orders by payment type



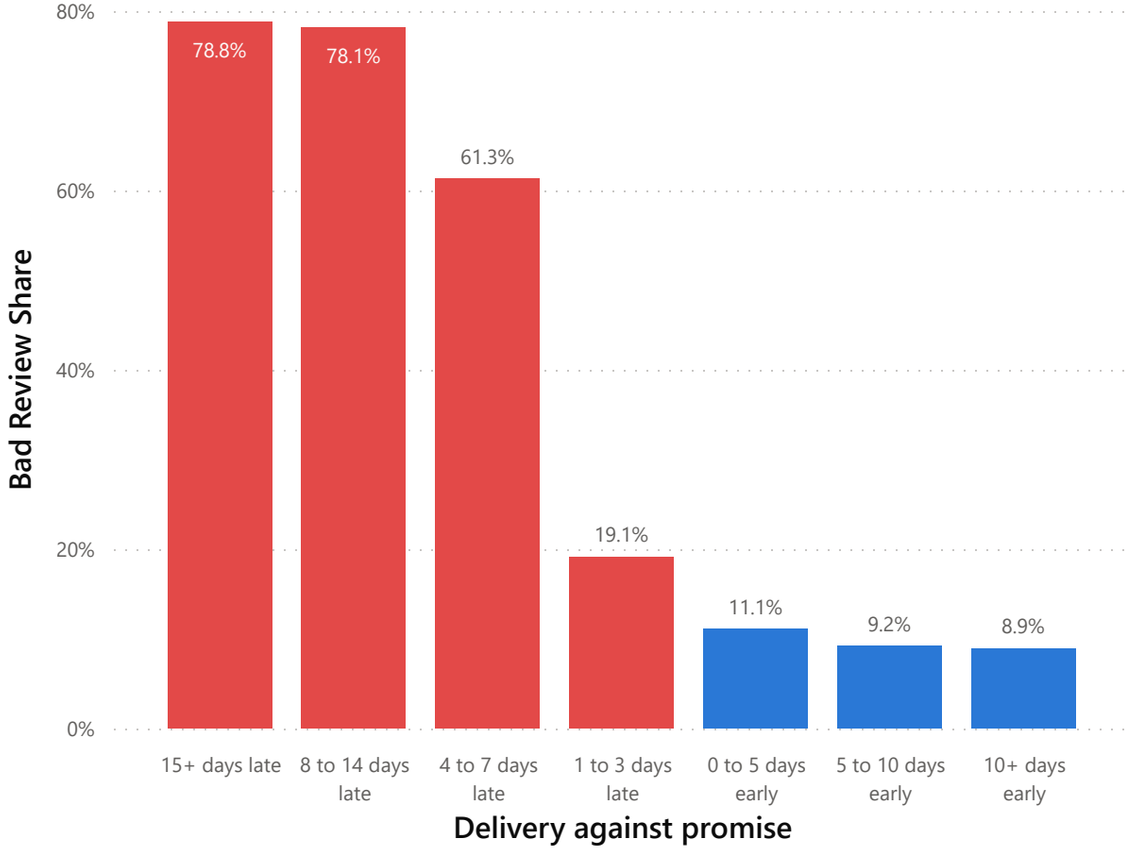
Delivery and reviews

Late parcels get bad reviews: past three days late, most customers leave one or two stars.

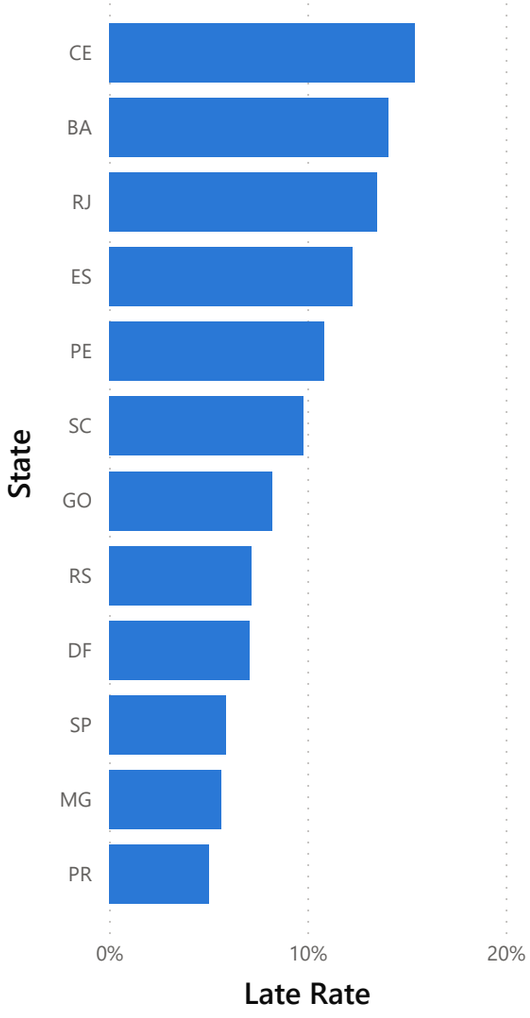
8.1%
Late Rate

13.8%
Bad Review Share

Share of 1 and 2-star reviews by delivery timing



Late rate by state (1,000+ orders)



Delivery by state

State	Delivered Orders	Late Rate	Average Delivery Days
CE	1272	15.4%	21.2
BA	3255	14.0%	19.3
RJ	12308	13.5%	15.3
ES	1991	12.3%	15.8
PE	1588	10.8%	18.4
SC	3534	9.8%	14.9
GO	1952	8.2%	15.6
RS	5327	7.2%	15.3
DF	2076	7.0%	12.9
SP	40396	5.9%	8.7
MG	11322	5.6%	12.0
PR	4905	5.0%	12.0

Cohorts and repeat buying

Under 1% of any cohort buys again in a later month.

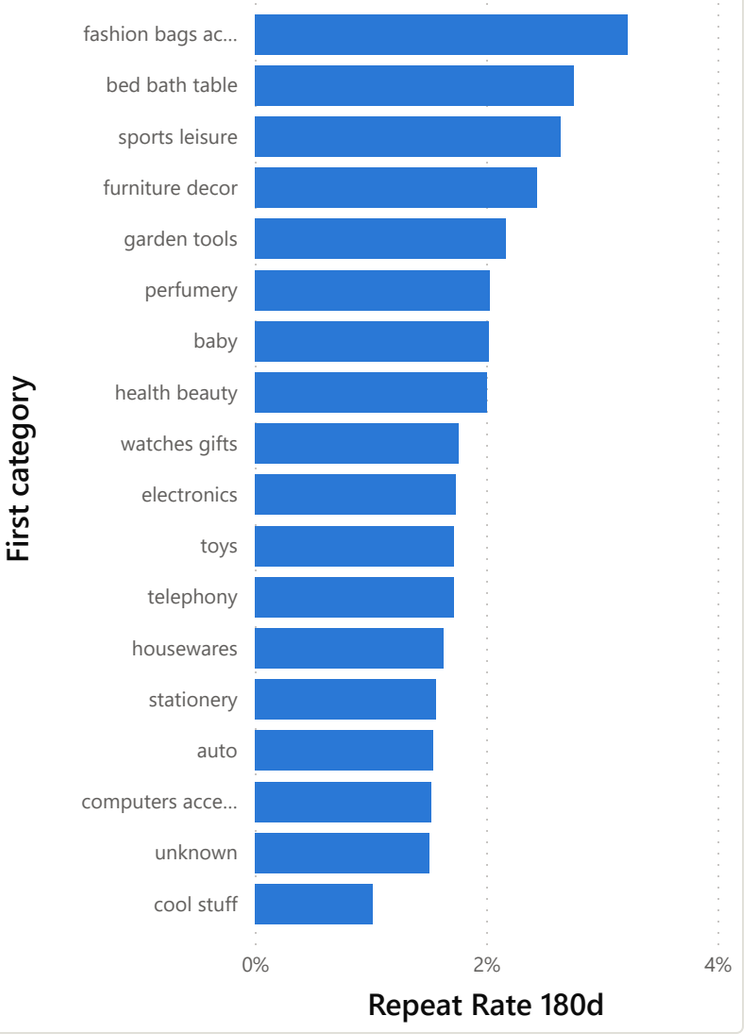
2.17%

Repeat Rate Ever

Share of each first-purchase cohort buying again, months 1 to 12

First-purchase month	1	2	3	4	5	6	7	8	9	10	11	12
January 2017	0.4%	0.3%	0.1%	0.4%	0.1%	0.4%	0.1%	0.1%	0.0%	0.4%	0.1%	0.7%
February 2017	0.2%	0.3%	0.1%	0.4%	0.1%	0.2%	0.2%	0.1%	0.2%	0.1%	0.3%	0.1%
March 2017	0.5%	0.4%	0.4%	0.4%	0.2%	0.2%	0.3%	0.4%	0.1%	0.4%	0.1%	0.2%
April 2017	0.6%	0.2%	0.2%	0.3%	0.3%	0.3%	0.3%	0.3%	0.2%	0.3%	0.1%	0.1%
May 2017	0.5%	0.5%	0.4%	0.3%	0.3%	0.4%	0.1%	0.3%	0.3%	0.3%	0.3%	0.3%
June 2017	0.5%	0.4%	0.4%	0.3%	0.4%	0.4%	0.2%	0.1%	0.2%	0.3%	0.3%	0.2%
July 2017	0.5%	0.3%	0.2%	0.3%	0.2%	0.3%	0.1%	0.2%	0.3%	0.2%	0.3%	0.1%
August 2017	0.7%	0.3%	0.3%	0.4%	0.5%	0.3%	0.3%	0.1%	0.1%	0.2%	0.2%	0.1%
September 2017	0.7%	0.5%	0.3%	0.4%	0.2%	0.2%	0.2%	0.3%	0.2%	0.3%	0.1%	
October 2017	0.7%	0.3%	0.1%	0.2%	0.2%	0.2%	0.4%	0.3%	0.2%	0.2%		
November 2017	0.6%	0.4%	0.2%	0.2%	0.2%	0.1%	0.2%	0.1%	0.1%			
December 2017	0.2%	0.3%	0.3%	0.3%	0.2%	0.2%	0.0%	0.2%				
January 2018	0.3%	0.4%	0.3%	0.3%	0.2%	0.2%	0.2%					
February 2018	0.4%	0.4%	0.3%	0.3%	0.2%	0.2%						
March 2018	0.4%	0.3%	0.3%	0.1%	0.1%							
April 2018	0.6%	0.3%	0.2%	0.1%								
May 2018	0.5%	0.3%	0.2%									
June 2018	0.4%	0.3%										
July 2018	0.5%											

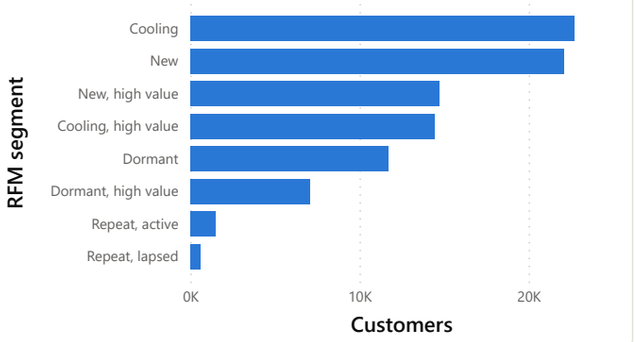
180-day repeat rate by first category (1,000+ customers)



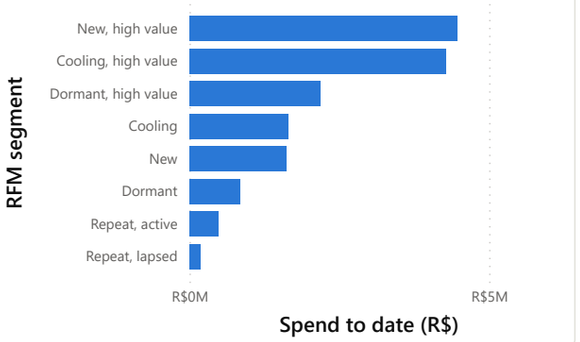
Segments and targeting

The model's top 10% returns at 1.8 times the average. That group is the target for a second-order test.

Customers by RFM segment

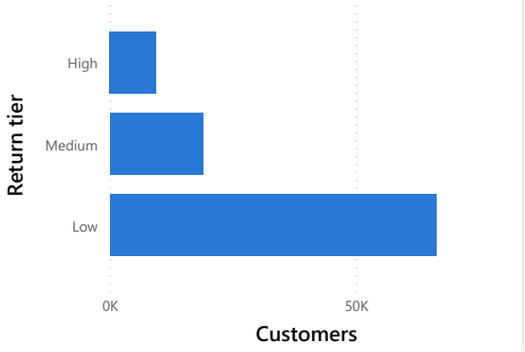


Total spend by RFM segment



Out-of-time test (notebook 04): the model's top 10% returned at 3.2% against 1.8% on average. Tiers shown here for older customers are in-sample and look better than that.

Customers by return tier



Action list: recent customers in the High tier, highest scores first

Customer	State	First category	Spend to date (R$)	Chance of returning in 180 days
f6337ff8052f9103de1a04029cb51b0a	SP	electronics	R$21.04	26.0%
d6dfb887a0dea041a5be2712dc856cde	SP	stationery	R$17.10	25.5%
30255bf74ce8634a06c8922ff0fa01cc	SP	electronics	R$21.04	25.3%
5a6cd8797286a15f0bf2fb97f024781c	SP	auto	R$21.39	25.3%
96a35e9dd5d9496820a8498a4bba0b99	SP	electronics	R$21.43	25.3%
d18fee94d33d23af9211025b59299edd	SP	telephony	R$25.29	25.3%
e5defab6ae60b311ca597ce8f5a66cf9	SP	drinks	R$25.67	25.3%
0294560ad384ad92c9aff267bbbd9406	SP	electronics	R$21.04	25.1%
5854a3b6b55a5a77abc18d59772cc3bc	SP	electronics	R$21.04	25.1%
94d15242e08e48c58b42c618dfc8b62d	SP	electronics	R$21.04	25.1%
c9c34696d5709b69f903d7d4807f0958	SP	health beauty	R$23.39	25.1%
21dbe8eabd00b34492a939c540e2b1a7	SP	sports leisure	R$14.29	23.5%
3a04ed9e9a9af78e22bd6b10e2e4d3dc	SP	electronics	R$20.29	20.5%